

Natuurlijke Taalverwerking (NTV) 2025/2026

Coordinators: Sandro Pezzelle & Wilker Aziz
Today's lecturer: Sandro Pezzelle

Hoorcollege HC1a

Welcome!

Today

- First part (13:00-13:45)
 - **Course organization & structure:** practical details including time, format, assignments and grades
- Second part (14:00-14:45)
 - **Content:** introduction to NLP tasks and applications

NTV Team

Sandro Pezzelle



- **Course Coordinator & Lecturer (Assistant Professor)**
- <https://sandropezzelle.github.io/>

Wilker Aziz



- Course Coordinator & Lecturer (Assistant Professor)
- <https://wilkeraziz.github.io/>

TA Team

- **Senior TAs:**
 - Frank Wildenburg
 - Mitchell Verhaar
- **TAs:** M. Verhaar, M. Bernardy, J. Bibo, S. Fris, J. Hutter, C. Maat, M. Oud, and J.M. Wormsbecher, Z. van den Berg

Organization

Overview

- **First edition!** NTV ‘replaces’ previous NTMI; starting next academic year, TTTV no longer there: NTV only NLP course
- 6 EC course: **168 hours of study** in 8 weeks
- 6 content weeks + 2 exam weeks (midterm + endterm)
- Time investment **around 20 hours/week**: 8 hours/week in class + 12 hours/week self-study

Platforms

- **Canvas** (our main tool; it contains all of the material)
- **Ed discussion** (asynchronous Q&A)
- **ANS** (exam-like exercises)
- **Jupyter Notebook** (tutorials)
- **Mentimeter & Wooclap** (live quizzes)

Study materials

- Lecture notes and/or textbook chapters
- Jupyter notebooks
- Lists of exercises and solutions
- Video tutorials, refreshers
- In-class quizzes (Mentimeter + Wooclap)

Learning activities

Hoorcolleges (twice a week)

- Purpose: present and discuss theory
- Preparation: 1-2 hours before each class
- Format: on site, recorded
 - recordings are meant as a complementary tool: prioritize attending classes!
- **Attendance is highly encouraged**
 - HCs are very interactive + **up to 6 in-class graded exercises (max 2 bonus points)**

***Laptopcollege* (once a week)**

- Purpose: connect theory and programming
- Format: on site, work in pairs
- Guided/supervised by one of the course's TAs
- Attendance is monitored and encouraged
- Contains graded and ungraded exercises
- Each tutorial requires 2-4 hours beyond the time in class

***Werkcollege* (once a week)**

- Purpose: solve exam-like exercises
- Format: on site
- Guided/supervised by one of the course's TAs
- Attendance is monitored and encouraged
- Each session requires about 2 hours beyond the time in class

Note on learning activities

- Our learning activities (HCs, LCs, WCs) are interconnected
- Consistently skipping or postponing any one of them will harm your performance and cause stress
- The best way to take this course is in real time

Self-study

- Reading time, ungraded exercises, graded homework, etc.
 - It is a core learning activity that should not be underestimated in this course
 - Reserve sufficient time for it (12h/week)
 - Live sessions do not replace self-study

Assessment

Homework component

- **Weekly programming exercises (20%)**
 - Graded by your TA. Best 5 out of 6 contribute to homework component
- **Weekly exam-like exercises (0%)**
 - These are ungraded, but very important for your learning. You can submit them on ANS and self-grade them with the model answers and the detailed rubrics. You can ask your TA for feedback.

Beyond graded assignments

- There is plenty of **ungraded exercises**; they help you learn and assess your knowledge, they are part of the study material and are just as likely to matter for an exam as anything else in the course
- They often expand beyond what is discussed in class, their solutions are shared in advance and you are welcome to ask questions about them at HCs or WCs

Deadlines

Beyond graded assignments

- Always on **Monday at 7pm (19:00) Amsterdam time**
 - e.g., February 9, 2026, at 7pm for assignment #1
- By default, late submissions are not graded
- Exceptions due to special circumstances are granted at our discretion based on input from the study advisor

Exam component

- Midterm (40%)
- Final (40%)
- * Resit replaces both exams (80%)

Grades

- Individual grades in a scale 0-10
- Course grade is weighted average of *exam component* (80%) and *homework component* (20%)
- Exam component must be 5.0 or more (if not > resit)
- Course grade rounded to nearest 0.5 point (6.7 mapped to 6.5)
 - except between 5-6, where we round to nearest point
- See <https://canvas.uva.nl/courses/56424/pages/grade>

All details about Organization

- Canvas/Modules/General Course Information
- https://canvas.uva.nl/courses/56424/modules#module_498038



Feedback & Contact

- When something is not going right, reach out, we hear you.
- Stay tuned for feedback session (*klankbordgroepen*) organised by the Program Committee AI (OC-AI)
- Reach out to TAs about LCs and WCs
- Reach out to senior TAs about organization
- Reach out to Sandro/Wilker about HCs and coordination

OC-AI

- The AI's programme committee (OC-AI) helps students and teachers address problems with the quality of the programme. They are eager to hear from you about this and any other course in the BKI and Master AI.
- Email: ocai-science@uva.nl

Questions?

Test on course organization



After the break

Introduction to **natural language processing (NLP)**

Natural language processing

Overview

- **What is natural language processing (NLP)**
- Why is NLP important
- What makes NLP difficult
- The statistical approach to NLP

What is NLP

- NLP is the branch of artificial intelligence (AI) that enables computers to automatically comprehend, generate, and manipulate human language
- NLP applies to both speech and written text and all human languages (English, Dutch, Italian, etc.)
- More recently, NLP models have extended beyond language, encompassing other modalities such as images, videos, code, etc.

NLP in 2026

- NLP is very popular these days, as it powers many user-facing, real-world language technology (based on large language models; LLMs) such as ChatGPT, Claude, Gemini, etc.
- The use of NLP-based systems, such as LLMs, into complex work processes, such as AI agents, is shaping the current AI ‘revolution’.

An interdisciplinary field



**NLP has to do
with human
natural language**

An interdisciplinary field

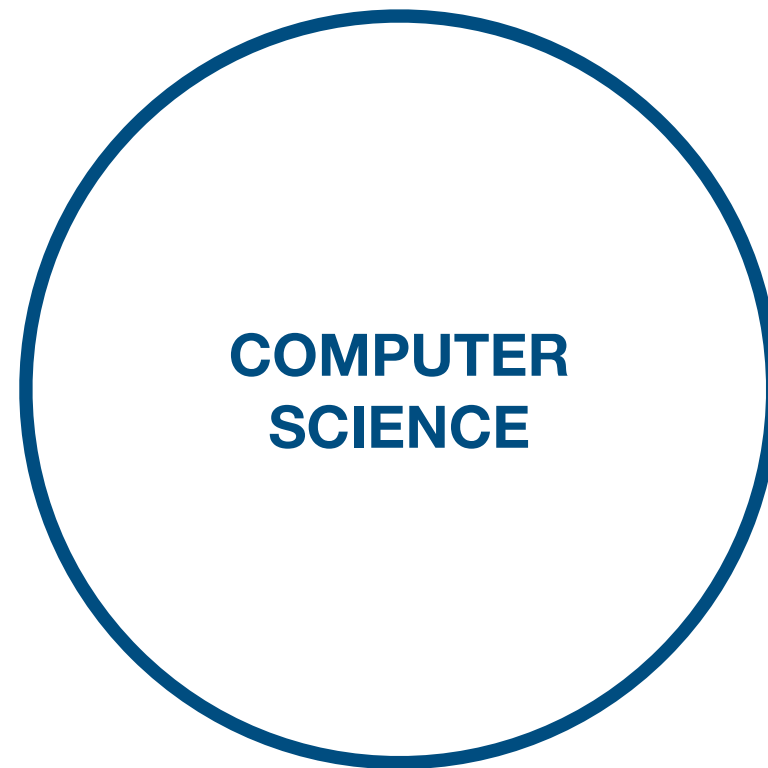


An interdisciplinary field

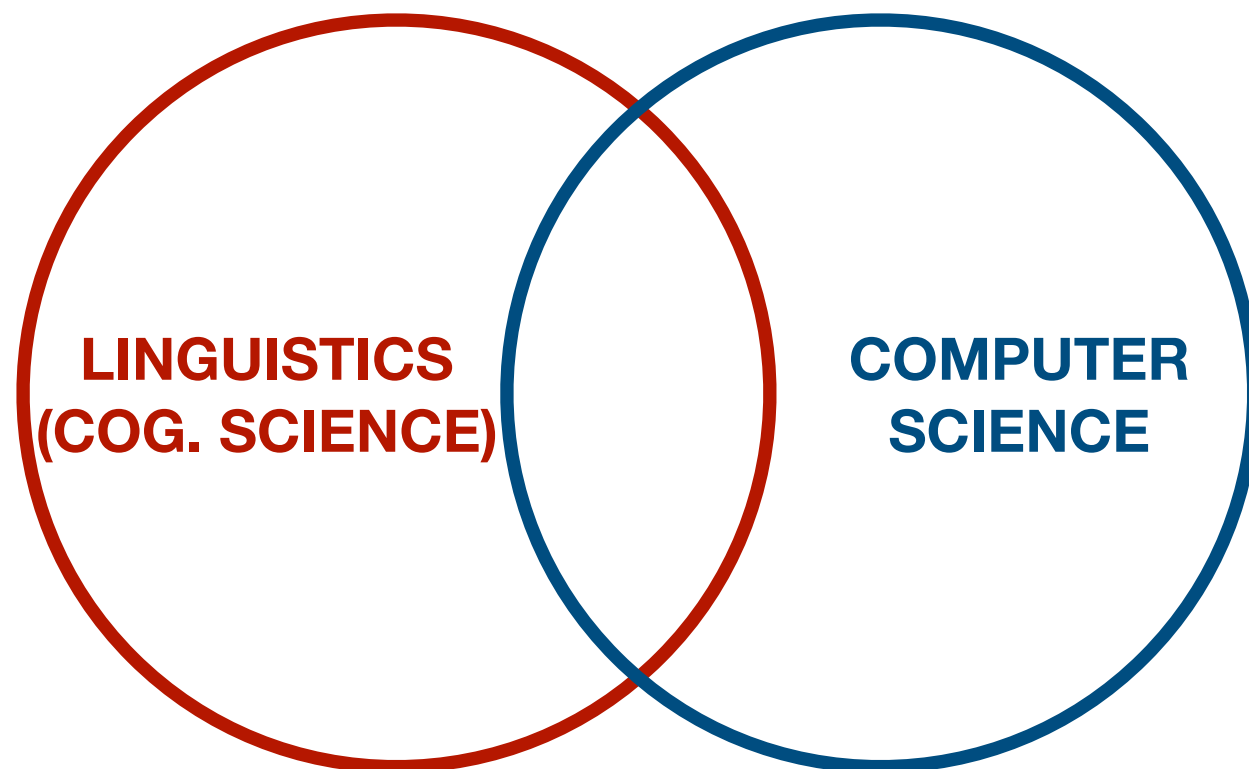


**NLP has to do with
automatic language
understanding &
generation by
computers**

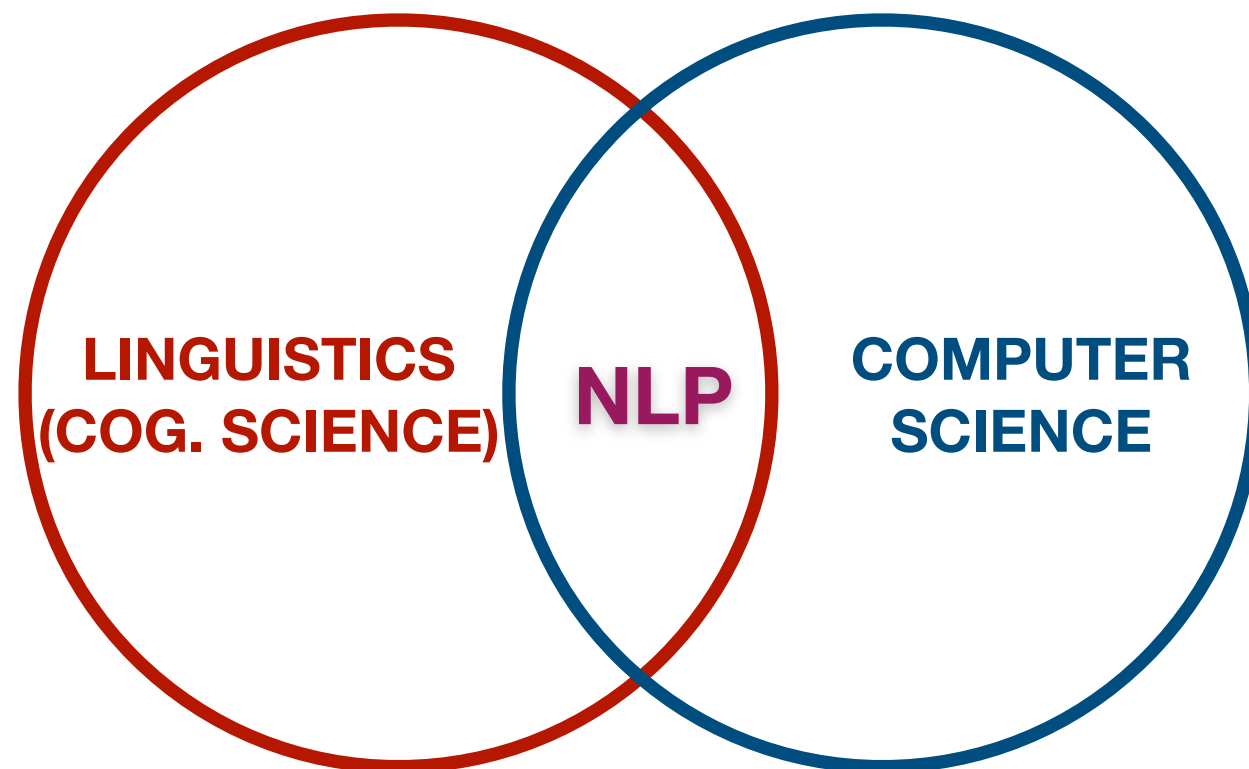
An interdisciplinary field



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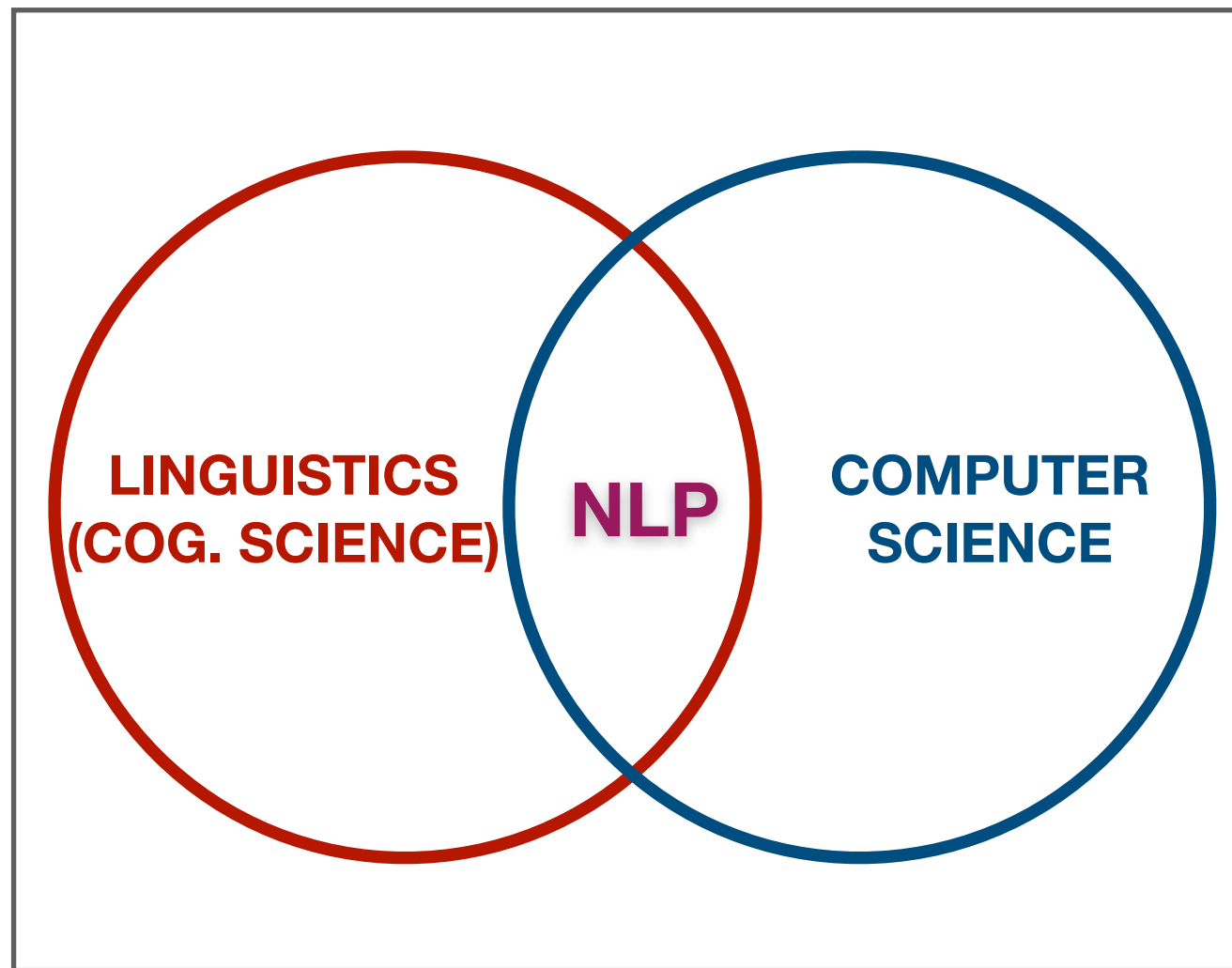


An interdisciplinary field



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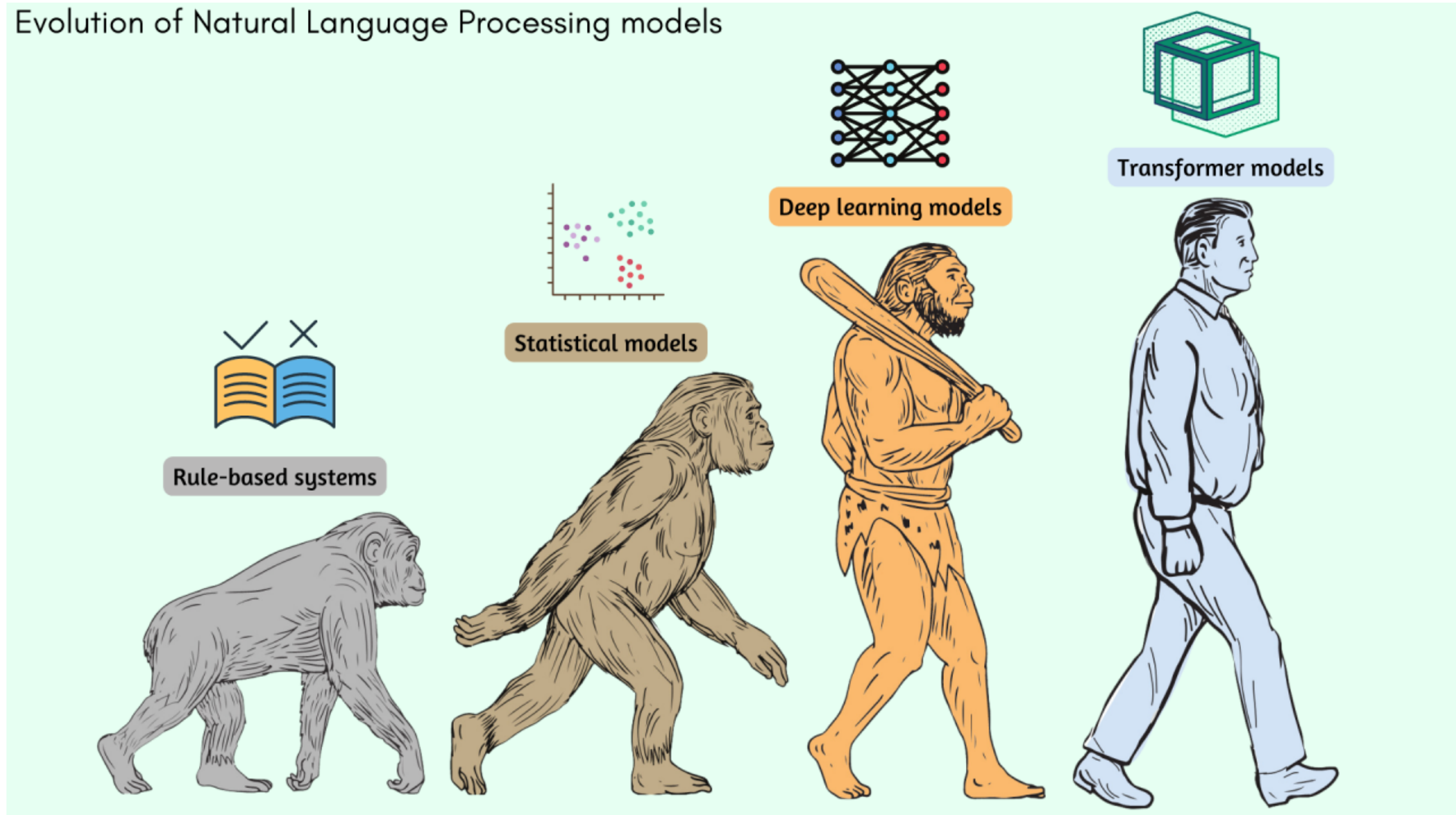
ARTIFICIAL INTELLIGENCE



From symbolic to ML-based NLP

- NLP—and AI at large—started in the 1950s in the form of rule-based systems, symbolic logic, and “expert systems” following **hard-coded rules** (seen in TTTV!)
- From the 1980s: statistical, probabilistic approach **learning from data** > machine learning-based NLP (this course!)

From symbolic to ML-based NLP



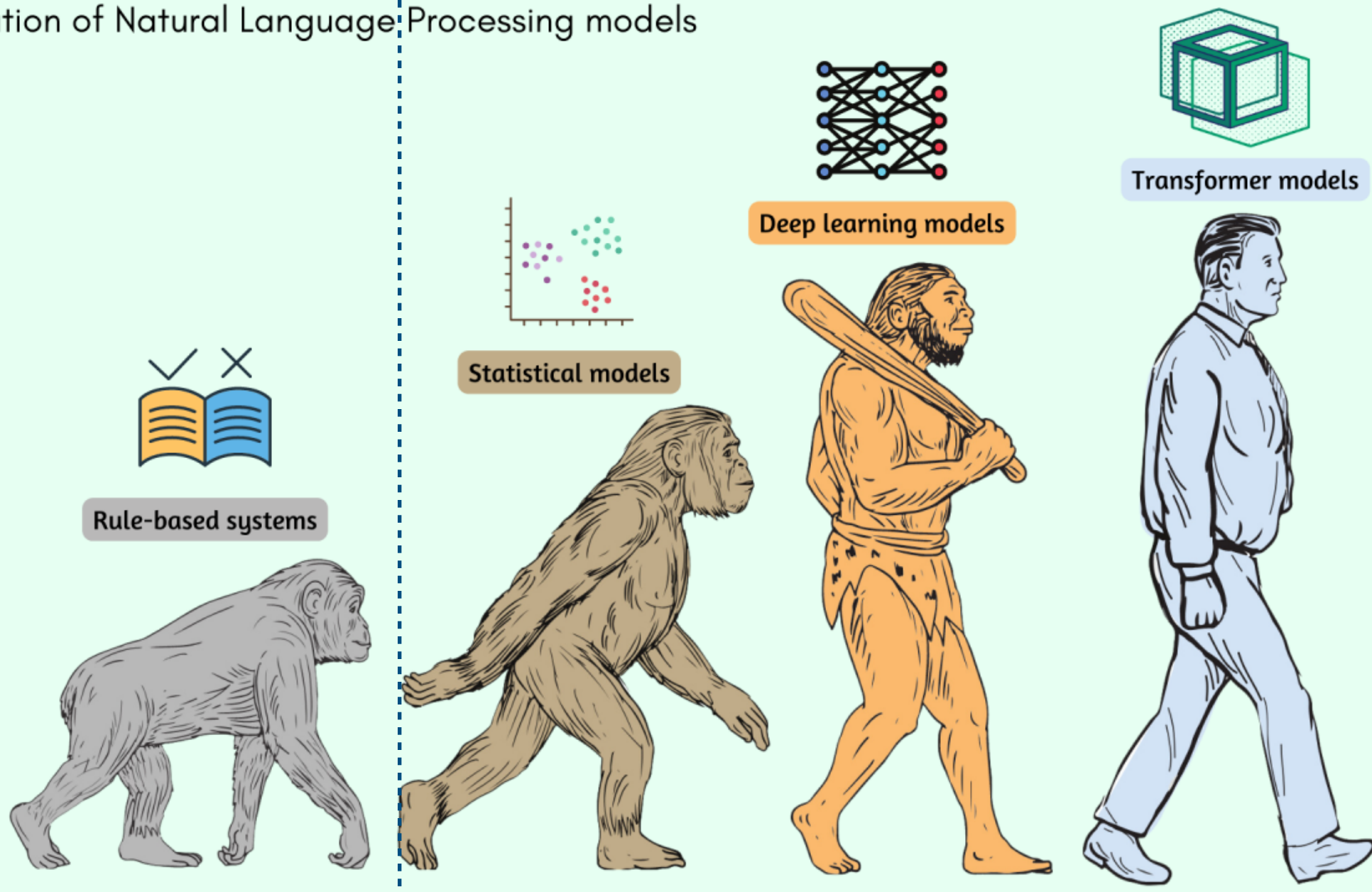
<https://www.linkedin.com/pulse/from-rulesets-transformers-journey-through-evolution-sota-yeddula/>

From symbolic to ML-based NLP

symbolic

data-driven

Evolution of Natural Language Processing models



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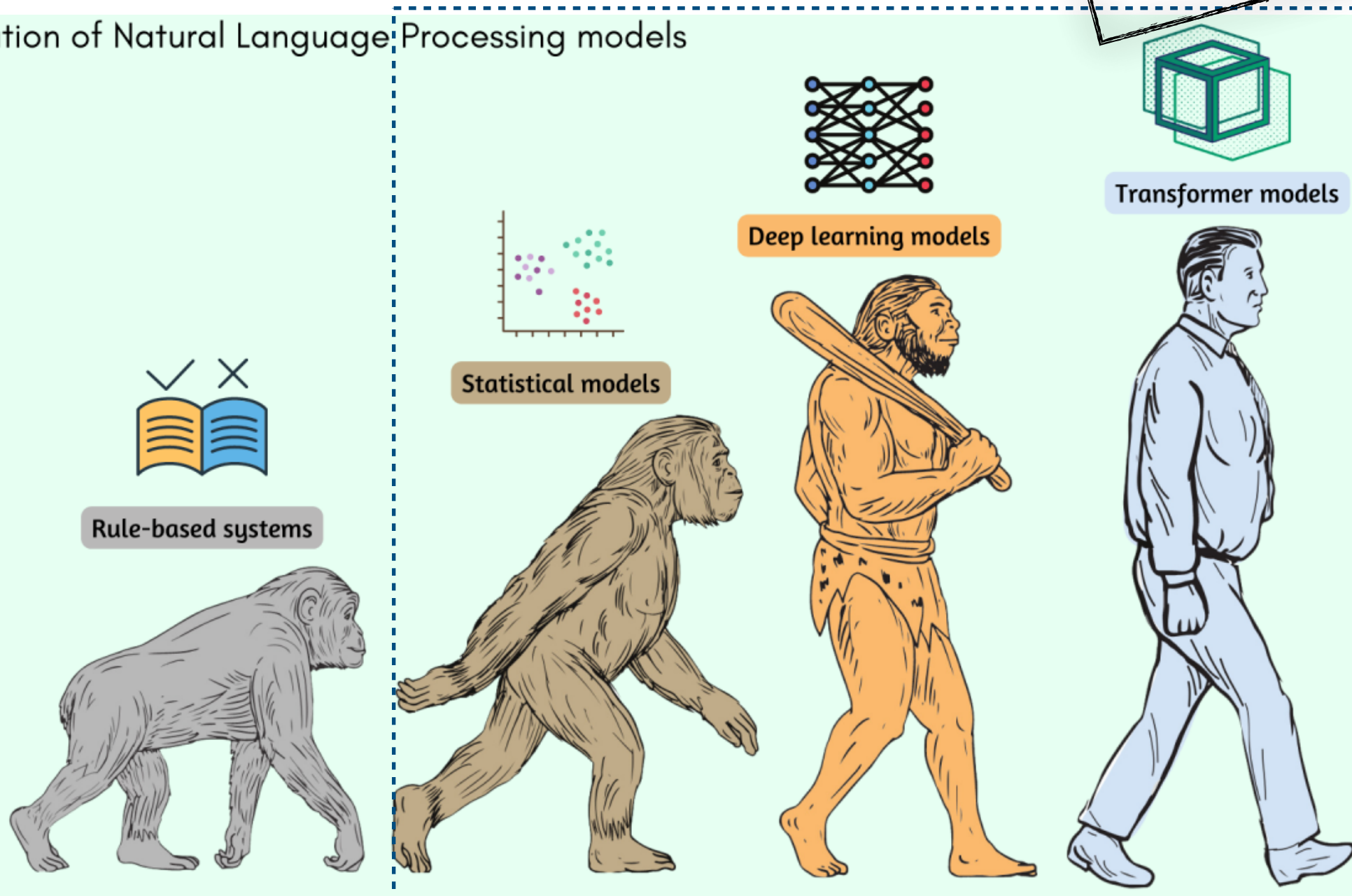
From symbolic to ML-based NLP

in this course!

symbolic

data-driven

Evolution of Natural Language Processing models



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A third approach

- **Neurosymbolic AI:**
 - hybrid approach that merges machine/deep learning with symbolic AI to create more robust, interpretable, and efficient AI systems
 - goal: mimick human-like thinking by combining pattern recognition with logical reasoning > better generalization with less data: less hallucinations, more transparency

A third approach

Symbolic and Neural AI,
3rd year course

- **Neurosymbolic AI:**
 - hybrid approach that merges machine/deep learning with symbolic AI to create more robust, interpretable, and efficient AI systems
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Overview

- What is natural language processing (NLP)
- **Why is NLP important**
- What makes NLP difficult
- The statistical approach to NLP

Language is everywhere

We use it to:

- communicate, i.e., talk about any sort of things
- store and transmit knowledge
- convey facts and give instructions
- express our feelings and imagination
- entertain ourselves (literature, theatre, movies)

A lot of useful applications

NLP is useful for:

- Information retrieval, summarization, and extraction
- Recommender systems (books, movies, restaurants)
- Translation
- Speech recognition
- Dialogue systems (chatbots)
- Multimodal applications: real-time video description, caption and subtitles generation, text-to-image generation, etc.

Some real-life examples

You probably use some of this technology daily:

- Info retrieval & summarization: 'AI mode' Google search (Gemini), review summaries of Amazon products, etc.
- Recommender systems: Amazon conversational assistant (Rufus)
- Translation: Google translate
- Speech recognition: Siri, Hey Google
- Dialogue systems: ChatGPT, Claude, Gemini...
- Multimodal applications: image generation w/ GPT or Qwen-Omni

NLP: Engineering side

Build models that power technological applications, e.g., the ones in the previous slide

NLP: Scientific side

Build models that help us understand more about human language: how do we (humans) learn language, what makes languages learnable, what do we use language for, what language tells us about social norms, etc.

In this course

- **We mostly focus on engineering aspects**
- ... but draw insights from scientific knowledge
- ... and reflect upon the impact of technology on society!

Overview

- What is natural language processing (NLP)
- Why is NLP important
- **What makes NLP difficult**
- The statistical approach to NLP

Human-level text analysis and generation

- require a language faculty
- involve semantic and pragmatic inference
- exploit world knowledge
- have purpose and intent
- are highly productive and ambiguous

Let's move to Menti!

Overview

- What is natural language processing (NLP)
- Why is NLP important
- What makes NLP difficult
- **The statistical approach to NLP**

Where should we start?

We will have to begin by accepting something very unintuitive:
language is pretty much random!

But, is language really random?

Well, *we* **do not** use language at random! We use language for a purpose and are aware of it

Instead, computers lack

- language faculty
- background knowledge
- self-driven purpose and intent

For a computer programme, language use is stochastic

- (having a random probability distribution or pattern that may be analysed statistically but may not be predicted precisely)
- In NLP, our models of text analysis and/or generation are all but very poor approximations of the language faculty, they lack background knowledge, and have no self-driven purpose.
- Whatever these models cannot explain is treated as randomness.

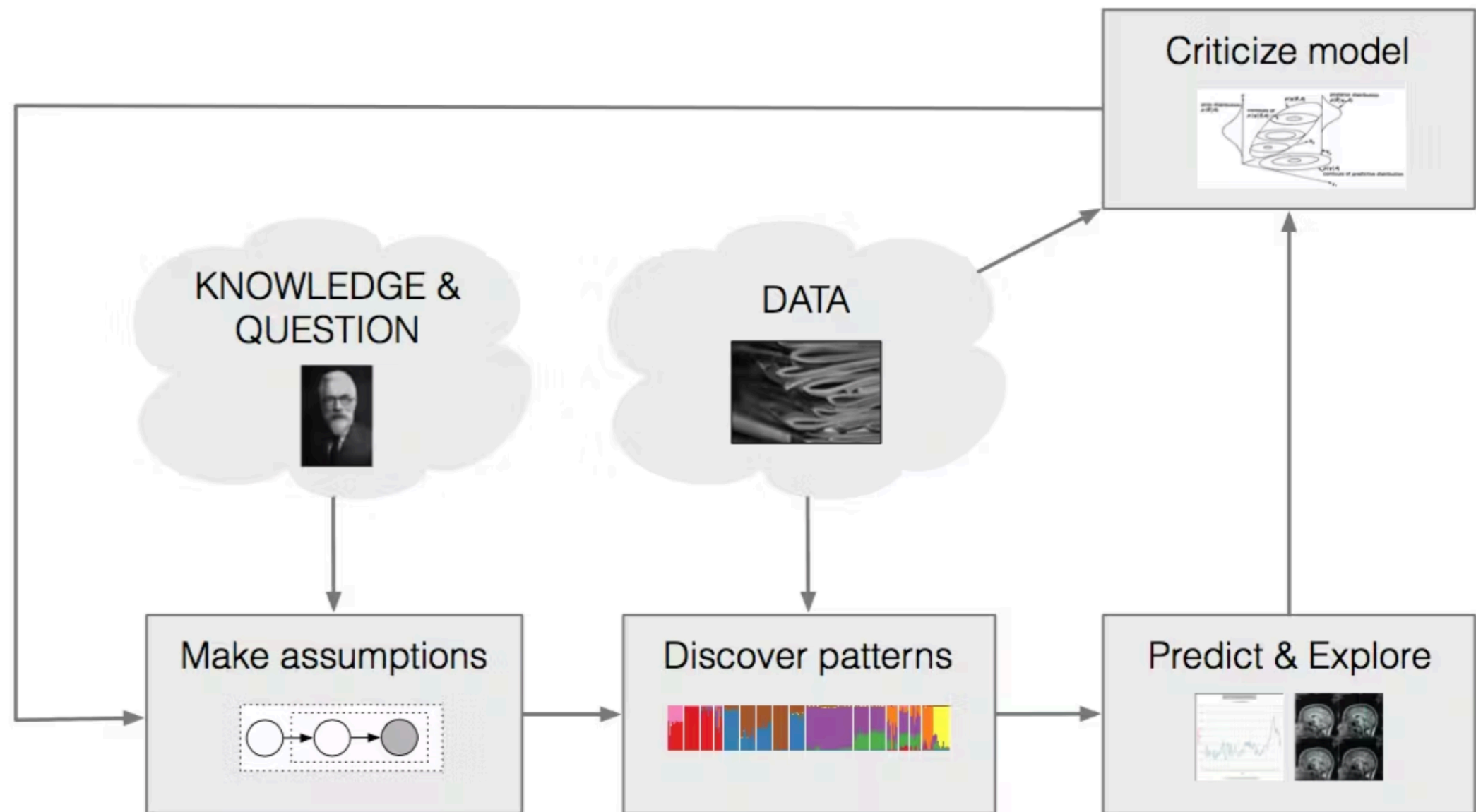
Randomness is not arbitrariness

- But randomness does not imply *arbitrariness*.
- Language is not arbitrary. Rather, it is **non-deterministic**.

Randomness can be constrained by available information

- Probability theory tells us how to represent and manipulate uncertainty.
- Statistics gives us techniques to model random experiments while minimising our uncertainty as much as possible.

The statistical approach or probabilistic pipeline



In NTV you will learn

- to design probabilistic models
- and perform statistical inference and estimation
- in order discover patterns in text
- with the goal of analyzing and generating linguistic data

What we cover

1. Statistics for NLP
2. Text classifiers
3. Feature learning
4. Language modelling
5. Sequence-2-sequence modelling
6. Self-supervised Pretraining